Prepared by Group 10

Activity 2 Task 2

Data Mining

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Q1 – What is Knowledge Discovery in Databases (KDD)?

Knowledge Discovery in Databases(KDD) is an automatic, exploratory analysis and modeling of large data repositories. KDD is the organized process of identifying valid, novel, useful, and understandable patterns from large and complex data sets.

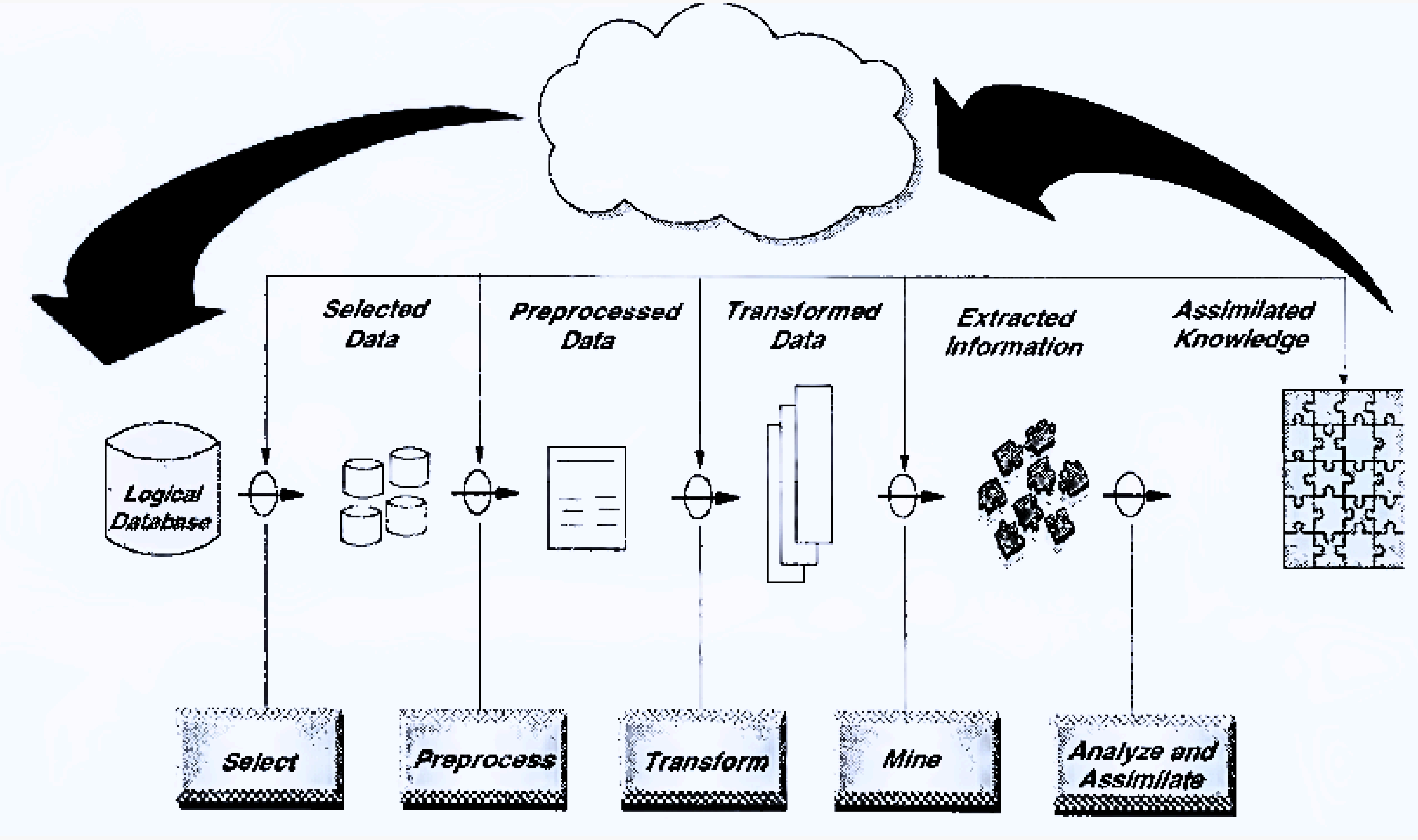
source: https://www.researchgate.net/publication/225835494_Chapter_1_-_Introduction_to_Knowledge_Discovery_in_Databases

Alternative definition

Knowledge discover in databases (KDD) (1980s) refers to the entire process of find patterns and similarities in raw data.

Source: Discovering Data Mining: From Concept to Implementation (IBM Books), Prentice Hall; Prentice-Hall International, Upper Saddle River, N.J, New Jersey, 1998. Peter Cabena; IBM (Armonk, N.Y.). International Technical Support Organization.

Q2 – What are the steps in the KDD process? (Include a figure/diagram)



The Knowledge Discovery in Databases (KDD) process involves several key steps:

- data selection
- data cleaning and preprocessing
- data transformation
- data mining
- pattern evaluation
- knowledge representation.

Source: Discovering Data Mining: From Concept to Implementation (IBM Books), Prentice Hall; Prentice-Hall International, Upper Saddle River, N.J, New Jersey, 1998. Peter Cabena; IBM (Armonk, N.Y.). International Technical Support Organization.

Q3 – How does KDD differ from Data Mining? (KDD vs. Data Mining)

Aspect	KDD (Knowledge Discovery in Databases)	Data Mining
Definition	KDD is the overall process of discovering useful knowledge using interesting patterns in database.	Data mining is the steps in applying mining algorithms during the process of knowledge discovery in database.
Scope	KDD is broad, covering multiple disciplines including statistics, artificial intelligence, machine learning, database science and information retrieval	Data mining is focused on specific algorithmic techniques to et different results.
Process & Techniques	• data selection • data cleaning and preprocessing • data transformation • data mining • pattern evaluation • knowledge representation	• Association analysis • Classification analysis • Clustering analysis
Outcome	Knowledge that can be interpreted and applied for business or research insights.	Identified patterns, associations, and predictive models.

sources: Data Mining Lecture notes Module 1b

Q4 – What is Data and Insights?

Data refers to a collection of discrete or continuous values that convey information, collected using techniques such as measurement, observation, query, or analysis, and are typically represented as numbers or characters that may be further processed.

Insights, on the other hand, are the deep understandings or realizations derived from analyzing and interpreting data. They represent the ability to perceive clearly or deeply into a situation or problem, often leading to actionable conclusions. For instance, analyzing the temperature data might reveal a pattern indicating a heatwave, which is an insight.

source: https://en.wikipedia.org

Alternative definition

Data is a collection of facts, representing measurements or descriptions of a specific situation. These facts can be in the form of numbers, symbols or words and are typically stored digitally.

Insight refers to an analyst or business user discovering a pattern in data or a relationship between variables that they didn't previously know existed.

source: https://www.qlik.com/us/data-analytics/data-insights

Q5 – What is the difference between data and insights?

Aspects	Data	Insights
Definition	Raw, unprocessed facts and figures.	Deep understanding and actionable knowledge derived from analyzing information.
Relationship	can be converted into insights through data processing and information analysis	Gathered by doing data processing and information analysis iteratively involving examining the information to identify trends, correlations, and anomalies to convert data to information, informations to insights
Decisions	Data-driven decisions rely on the availability of accurate and comprehensive data. By basing decisions on raw data, businesses can ensure objectivity and factual accuracy. Data-driven decisions are foundational, providing the necessary facts to support further analysis.	Insight-driven decisions are the pinnacle of the decision-making hierarchy. These decisions are strategic and forward-thinking, informed by deep insights that reveal underlying trends and causal relationships. Insight-driven decisions help businesses anticipate market changes, optimize strategies, and achieve competitive advantages.

sources: https://www.market-xcel.com/blogs/data-information-and-insight-understanding-their-essential-differences-interactions

Q6 – What is CRISP-DM (Cross Industry Standard Process for Data Mining)?

CRISP-DM is a popular analytics lifecycle that acts as a structured framework for planning and executing data mining projects. The process consists of six major phases:

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modeling
5. Evaluation
6. Deployment

source: https://medium.com/@shawn.chumbar/the-crisp-dm-process-a-comprehensive-guide-4d893aecb151

Q7 – How does Information Retrieval (IR) differ from Data Mining?

Aspect	Information Retrieval (IR)	Data Mining (DM)
Definition	a process that facilitates the effective and efficient retrieval of relevant information from large collections of unstructured or semi-structured data.	the process of sorting through large data sets to identify patterns and relationships that can help solve business problems through data analysis.
How does it works	Retrieves and ranks relevant documents or information based on user queries.	Extracts hidden patterns, trends, or insights from structured and unstructured data.
Data Type	Works with unstructured or semi-structured data (text, images, videos).	Deals mainly with structured or numerical data (databases, spreadsheets).
Techniques Used	Uses statistical, linguistic, and heuristic methods such as NLP, indexing, and ranking.	Relies on machine learning, artificial intelligence, and mathematical methods such as clustering and classification.
Applications	Search engines, recommender systems, text summarization, sentiment analysis.	Market analysis, fraud detection, medical predictions, scientific data classification.
Challenges	Handling ambiguity, data heterogeneity, usability, and ethical concerns (privacy, bias).	Data preprocessing, feature selection, algorithm selection, result interpretation.

source: https://www.linkedin.com/advice/3/how-do-you-apply-information-retrieval-vs

Q8 – What is Machine Learning (ML)?

Machine learning (ML) is a discipline of artificial intelligence (AI) that provides machines with the ability to automatically learn from data and past experiences while identifying patterns to make predictions with minimal human intervention.

Machine learning methods enable computers to operate autonomously without explicit programming. ML applications are fed with new data, and they can independently learn, grow, develop, and adapt.

source: https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-ml/#:~:text=Machine%20learning%20(ML)%20is%20a,predictions%20with%20minimal%20human%20intervention.

Q8 – What is Machine Learning (ML)? (Alternative definitions)

In machine learning, a computer learns from example data how to perform tasks. We know that if we give more experiences with a defined task to a machine, its performance improves. For example, let suppose that we want an email client to classify emails as spam or not. The experience E in this case should be a set of emails already classified as spam or not. The Task T perform is to classify automatically new emails. The performance P that should increase is the accuracy rate of the classification made by the machine on a set of new emails.

source: R. Farhat, Y. Mourali, M. Jemni and H. Ezzedine, "An overview of Machine Learning Technologies and their use in E-learning," 2020 International Multi-Conference on: “Organization of Knowledge and Advanced Technologies” (OCTA), Tunis, Tunisia, 2020, pp. 1-4, doi: 10.1109/OCTA49274.2020.9151758.

Q9 – What are the main types of ML?

Supervised learning — where the algorithm generates a function that maps inputs to desired outputs. One standard formulation of the supervised learning task is the classification problem: the learner is required to learn (to approximate the behavior of) a function which maps a vector into one of several classes by looking at several input-output examples of the function.

Unsupervised learning — which models a set of inputs: labeled examples are not available.

Semi-supervised learning -—- which combines both labeled and unlabeled examples to generate an appropriate function or classifier.

Reinforcement learning —— where the algorithm learns a policy of how to act given an observation of the world. Every action has some impact in the environment, and the environment provides feedback that guides the learning algorithm.

Source: Zhang, Y. (2010). New Advances in Machine Learning. IntechOpen. Retrieved from https://books.google.com.my/books?id=XAqhDwAAQBAJ.

Q10 – How is DM different from ML (DM vs. ML)?

Aspect	Data Mining	Machine Learning
Definition	The process of examining large datasets to find hidden patterns, relationships and insights.	A field of study that enables computers to learn from data and improve their performance over time without being programmed in that way.
Purpose	Designed to extract rules and patterns from a large amounts of data to help with decision-making.	Focuses on developing algorithms that allow systems to learn from data, make predictions and adapt to new situations.
Techniques Used	Use statistical methods, pattern recognition and clustering to analyze data.	Use algorithms such as neural networks, decision trees and support vector machines to learn from data.
Human Involvement	Requires significant human intervention to select, preprocess and interpret the results.	Once algorithms are defined, minimal human intervention is needed as the system learns autonomously.
Outcome	Discovers previously unknown patterns and relationships in data.	Produces models that can predict outcomes and improve performance based on new data.

source: https://www.simplilearn.com/data-mining-vs-machine-learning-article?

Q11 – What is Artificial Intelligence (AI)?

Artificial intelligence (AI) refers to the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, perception, and decision-making. It is a field of research in computer science that develops and studies methods and software that enable machines to perceive their environment and use learning and intelligence to take actions that maximize their chances of achieving defined goals.

source: https://en.wikipedia.org/wiki/Artificial_intelligence

Alternative definition

Artificial intelligence is a field of science concerned with building computers and machines that can reason, learn, and act in such a way that would normally require human intelligence or that involves data whose scale exceeds what humans can analyze.

source: https://cloud.google.com/learn/what-is-artificial-intelligence

Q12 – What is the difference between ML and AI (ML vs. AI)?

Aspect	Artificial Intelligence (AI)	Machine Learning (ML)
Definition	AI refers to the use of technologies to build machines and computers that have the ability to mimic cognitive functions associated with human intelligence, such as being able to see, understand, and respond to spoken or written language, analyze data, make recommendations, and more.	ML is a subset of artificial intelligence that automatically enables a machine or system to learn and improve from experience. Instead of explicit programming, machine learning uses algorithms to analyze large amounts of data, learn from the insights, and then make informed decisions.
Goal	The goal is to develop an intelligent system that can perform complex tasks.	The goal is to build machines that can learn from data to increase the accuracy of the output.
Approach	AI uses technologies in a system so that it mimics human decision-making.	ML uses self-learning algorithms to produce predictive models.
Data Utilization	AI works with all types of data: structured, semi-structured, and unstructured.	ML can only use structured and semi-structured data.
Learning Capability	AI systems use logic and decision trees to learn, reason, and self-correct.	ML systems rely on statistical models to learn and can self-correct when provided with new data.

source: https://cloud.google.com/learn/artificial-intelligence-vs-machine-learning#differences-between-ai-and-ml

Q13 – What are some real-world applications of AI and potential careers in this field?

Artificial intelligence (AI) is transforming various industries by enhancing efficiency and personalization. In healthcare, AI assists in diagnosing diseases, personalizing treatment plans, and predicting patient outcomes, leading to improved patient care. In the automotive sector, AI powers autonomous vehicles, enabling them to navigate, detect objects, and make decisions, thereby enhancing safety and efficiency. Additionally, in finance, AI is employed for fraud detection, risk assessment, and algorithmic trading, helping institutions analyze data patterns and make informed decisions.

Source: https://www.forbes.com/sites/bernardmarr/2023/05/10/15-amazing-real-world-applications-of-ai-everyone-should-know-about/

Q14 – Provide a figure that illustrates the relationships among KDD, DM, ML and AI.